

Great question! Let's expand the example with an additional feature and see how the tree construction changes.

Expanded Dataset with Two Features

Let's add a "Location" feature:

X	Location	y
1	A	15
2	A	20
3	B	25
4	B	30
5	B	35

Step 1: Calculate Root Node Statistics

All data in root node:

$$y = [15, 20, 25, 30, 35]$$

$$\text{Mean} = 25$$

$$\text{MSE}_{\text{root}} = 50 \text{ (same as before)}$$

Step 2: Evaluate All Possible Splits Across BOTH Features

Now we need to check splits for **both** X and Location:

Feature 1: Numerical Feature (X) Splits

Split 1: $X \leq 1.5$

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```
Left:  [1] → y = [15], MSE_left = 0
Right: [2,3,4,5] → y = [20,25,30,35], Mean = 27.5, MSE_right = 31.25
Weighted MSE = (1×0 + 4×31.25)/5 = 125/5 = 25
Gain = 50 - 25 = 25
```

Split 2: $X \leq 2.5$

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```
Left:  [1,2] → y = [15,20], Mean = 17.5, MSE_left = 6.25
Right: [3,4,5] → y = [25,30,35], Mean = 30, MSE_right = 16.67
Weighted MSE = (2×6.25 + 3×16.67)/5 = 62.5/5 = 12.5
Gain = 50 - 12.5 = 37.5
```

Split 3: $X \leq 3.5$

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```
Left:  [1,2,3] → y = [15,20,25], Mean = 20, MSE_left = 16.67
Right: [4,5] → y = [30,35], Mean = 32.5, MSE_right = 6.25
Weighted MSE = (3×16.67 + 2×6.25)/5 = 62.5/5 = 12.5
Gain = 50 - 12.5 = 37.5
```

Split 4: $X \leq 4.5$

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```
Left:  [1,2,3,4] → y = [15,20,25,30], Mean = 22.5, MSE_left = 31.25
Right: [5] → y = [35], MSE_right = 0
Weighted MSE = (4×31.25 + 1×0)/5 = 125/5 = 25
Gain = 50 - 25 = 25
```

Feature 2: Categorical Feature (Location) Splits

For categorical features, we consider all possible binary partitions:

Split 5: Location = A vs Location = B

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```
Location = A: [1,2] → y = [15,20], Mean = 17.5, MSE_left = 6.25
Location = B: [3,4,5] → y = [25,30,35], Mean = 30, MSE_right = 16.67
Weighted MSE = (2×6.25 + 3×16.67)/5 = 62.5/5 = 12.5
Gain = 50 - 12.5 = 37.5
```

Split 6: Location = A,B (other combinations don't make sense for binary split)

Step 3: Choose Best Split

Split	Feature	Threshold	Weighted MSE	Gain
1	X	≤ 1.5	25	25
2	X	≤ 2.5	12.5	37.5
3	X	≤ 3.5	12.5	37.5
4	X	≤ 4.5	25	25
5	Location	= A	12.5	37.5

Multiple best splits with gain = 37.5

Let's choose **Location = A** as our first split (it's equally good but might be more interpretable).

Step 4: Build Level 1 of Tree

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```
Root: Location = A?
├─ Yes (Left): [1,2] → Mean = 17.5, MSE = 6.25
└─ No (Right): [3,4,5] → Mean = 30, MSE = 16.67
```

Step 5: Split Left Child (Location = A region)

Left Child Data: Points [1,2], y = [15,20], MSE = 6.25

Evaluate possible splits:

Split on X: $X \leq 1.5$

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Left-Left: $[1] \rightarrow y = [15]$, $MSE = 0$

Left-Right: $[2] \rightarrow y = [20]$, $MSE = 0$

Weighted MSE = 0

Gain = $6.25 - 0 = 6.25$

Location is constant in this region, so no further splits on Location.

Best split: $X \leq 1.5$ with gain = 6.25

Step 6: Split Right Child (Location = B region)

Right Child Data: Points $[3,4,5]$, $y = [25,30,35]$, $MSE = 16.67$

Evaluate possible splits:

Split on X: $X \leq 3.5$

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Right-Left: $[3] \rightarrow y = [25]$, $MSE = 0$

Right-Right: $[4,5] \rightarrow y = [30,35]$, Mean = 32.5, $MSE = 6.25$

Weighted MSE = $(1 \times 0 + 2 \times 6.25) / 3 = 12.5 / 3 \approx 4.17$

Gain = $16.67 - 4.17 = 12.5$

Split on X: $X \leq 4.5$

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Right-Left: $[3,4] \rightarrow y = [25,30]$, Mean = 27.5, $MSE = 6.25$

Right-Right: $[5] \rightarrow y = [35]$, $MSE = 0$

Weighted MSE = $(2 \times 6.25 + 1 \times 0) / 3 = 12.5 / 3 \approx 4.17$

Gain = $16.67 - 4.17 = 12.5$

Location is constant in this region, so no further splits on Location.

Best split: $X \leq 3.5$ with gain = 12.5

Step 7: Final Tree Structure

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```
Root: Location = A?  
  |— Yes →  $X \leq 1.5$ ?  
    |— Yes: Predict 15 (Leaf)  
    |— No: Predict 20 (Leaf)  
  |— No →  $X \leq 3.5$ ?  
    |— Yes: Predict 25 (Leaf)  
    |— No: Predict 32.5 (Leaf)
```

Step 8: Verify Predictions

Let's test our tree:

X=1, Location=A: Location=A? Yes → $X \leq 1.5$? Yes → Predict 15 ✓

X=2, Location=A: Location=A? Yes → $X \leq 1.5$? No → Predict 20 ✓

X=3, Location=B: Location=A? No → $X \leq 3.5$? Yes → Predict 25 ✓

X=4, Location=B: Location=A? No → $X \leq 3.5$? No → Predict 32.5 ✓

X=5, Location=B: Location=A? No → $X \leq 3.5$? No → Predict 32.5 ✓

Step 9: Calculate Final MSE

Actual: [15, 20, 25, 30, 35]

Predicted: [15, 20, 25, 32.5, 32.5]

$$\begin{aligned} \text{MSE} &= [(15-15)^2 + (20-20)^2 + (25-25)^2 + (30-32.5)^2 + (35-32.5)^2] / 5 \\ &= [0 + 0 + 0 + 6.25 + 6.25] / 5 = 12.5/5 = 2.5 \end{aligned}$$

Key Differences with Multiple Features

1. More Split Options

With 1 feature: 4 possible splits

With 2 features: 6 possible splits (4 numerical + 2 categorical)

2. Feature Selection

The algorithm automatically picks the best feature at each step:

Root: Chose Location over X (both had same gain)

Children: Used the remaining feature (X)

3. Tree Structure Changes

Original tree (X only):

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$X \leq 2.5?$

├─ $X \leq 1.5?$ → Predict 15/20

└─ $X \leq 3.5?$ → Predict 25/32.5

New tree (X + Location):

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Location = A?

├─ $X \leq 1.5?$ → Predict 15/20

└─ $X \leq 3.5?$ → Predict 25/32.5

Same predictions, different interpretation!

4. Real-World Interpretation

The tree now tells a story:

First question: "Where is the property located?"

Then: "How big is it?"

This is much more interpretable than just using X alone!

Summary

Adding more features:

Increases computation (more splits to evaluate)

Improves interpretability (meaningful features)

May find better splits (different perspectives on the data)

Maintains same prediction quality in this case

The algorithm automatically handles both numerical and categorical features seamlessly!